

# Windpark Musterhöhe

Fictional onshore wind farm ·  $5 \times 6.3 \text{ MW} = 31.5 \text{ MW}$  · Hesse, Germany

Base case, as published on the website

Data as of 30 Sep 2026  
Engine 2.1.0  
Inputs 388d9b67b9e5fe67  
Calculator [igorsabodakha.com/wind-farm-calculator/](https://igorsabodakha.com/wind-farm-calculator/)

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## A model by Igor Sabodakha

Illustrative calculation — not investment, tax or legal advice. The wind farm is fictional. The results depend on the assumptions shown on slide 4.

At 4.79 ct/kWh the owners earn 3.1% a year — below an 8% cost of equity

Equity IRR

3.1%

KG, after the company's taxes; cost of equity 8%

Project IRR

3.7%

Unlevered, after tax (pre-tax 4.0%)

Equity NPV

-€16.1m

At 8%, to financial close

LCOE

7.25 ct/kWh

Real 2026 money (nominal 9.22 ct/kWh)

Min DSCR

1.44x

In 2030; covenant 1.05x, P90 1-yr 1.14x

Debt

€20.6m

34.8% of uses; equity €38.7m

Validity

✓ Valid: fully funded, covenant met

✓ CalculationOK · 9 checks

✓ FundingOK · 2 checks

✓ CovenantOK · 2 checks

✓ InputsOK · 6 checks

⚠ Model scopeWarning · 6 checks

⚠ Book equity stays positive (limits on distributions) — not met in 2043–2050 (slide 12)

Key messages

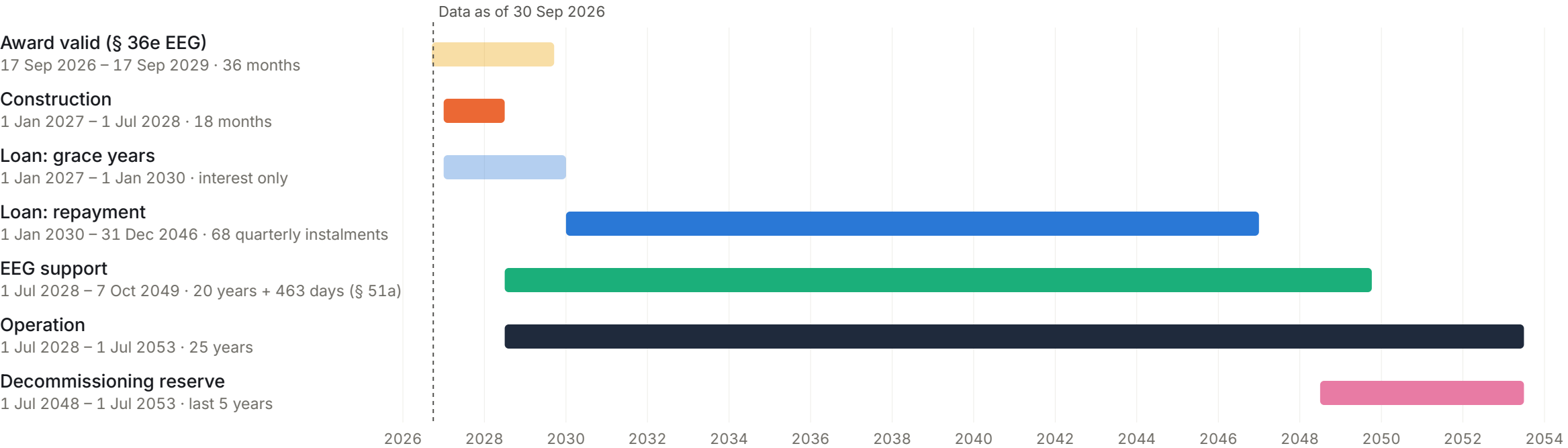
- The EEG floor (AW 6.30 ct/kWh) stays below the expected market value, so revenue comes from the market; the premium works as insurance and is paid in 15 years of the downside.
- The bank lends €20.6m (35% of uses), limited by the P90 (1-year) DSCR target on the EEG floor; the owners fund €38.7m.
- Debt lowers the owners' return: 3.1% against 3.7% for the project without debt, which earns less than the loan's 5.35%.
- Bid calculator: an 8% equity IRR needs an award of 7.41 ct/kWh — above the 7.25 ct/kWh ceiling.

Scenarios

|           | Equity IRR | Min DSCR |
|-----------|------------|----------|
| Base      | 3.1%       | 1.44x    |
| P90 1-yr  | 0.6%       | 1.14x    |
| P90 10-yr | 1.2%       | 1.20x    |
| Downside  | -2.1%      | 1.18x    |

The stress scenarios keep the base-case loan — the lender's view after financial close.

# Commissioning on 1 Jul 2028, 18 months after financial close; the loan is repaid by 31 Dec 2046



|                 |                                  |                    |                             |                  |                      |
|-----------------|----------------------------------|--------------------|-----------------------------|------------------|----------------------|
| Capacity        | 5 × 6.3 MW = 31.5 MW             | Hub height / rotor | 159 m / 165 m               | Site quality     | 68% → 2,439 h at P50 |
| Award announced | 17 Sep 2026; COD after 21 months | Financial close    | 1 Jan 2027 (valuation date) | First instalment | 31 Mar 2030          |
| Loan term       | 20 years, 3 grace years          | Legal form         | GmbH & Co. KG               | EEG support ends | 7 Oct 2049           |

Key assumptions

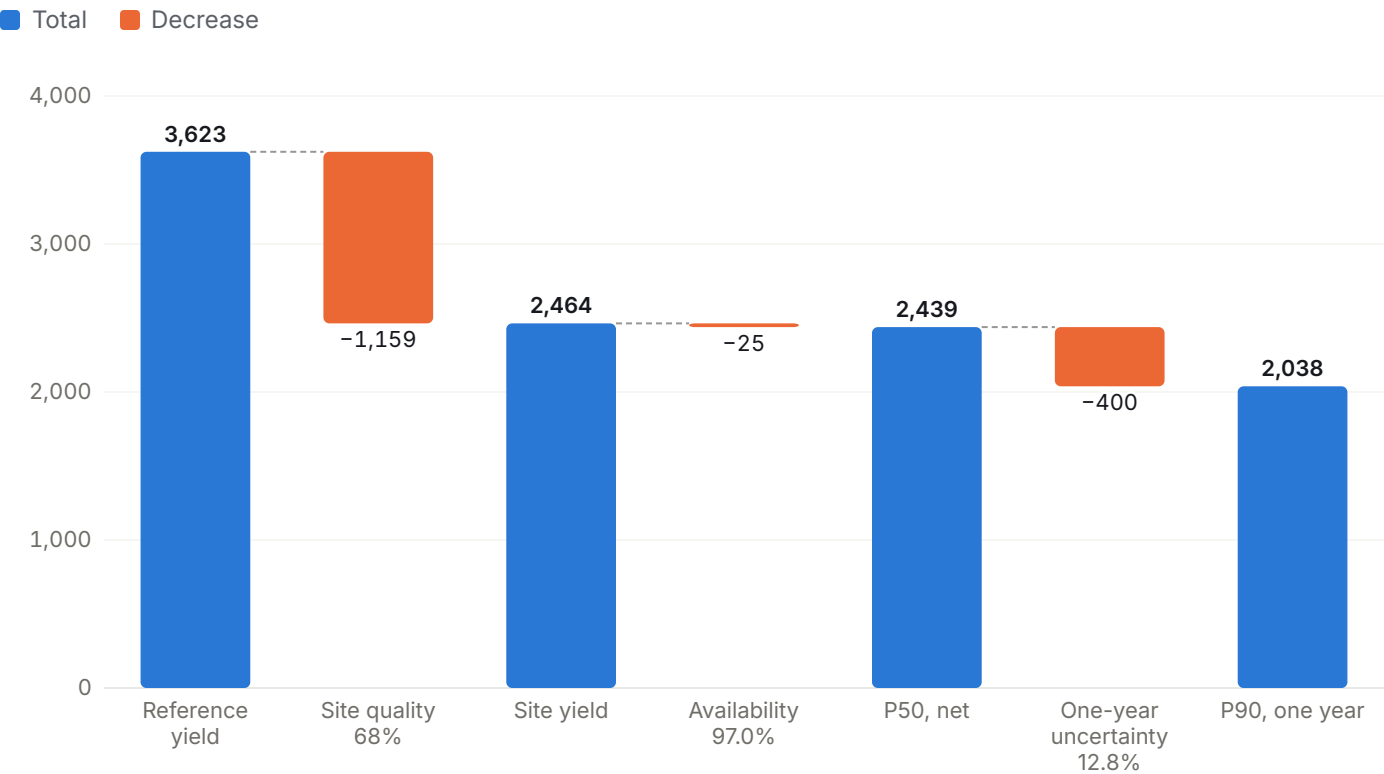
40 inputs drive the answer: 36 from public sources, 4 documented assumptions

| PROJECT                   |            |         | REVENUE                         |                     |             | FINANCING                          |                         |             |
|---------------------------|------------|---------|---------------------------------|---------------------|-------------|------------------------------------|-------------------------|-------------|
| Number of turbines        | 5          | A       | Award price<br>(Zuschlagswert)  | 4.79 ct/kWh         | S1          | Interest rate                      | 5.35%                   | S9          |
| Turbine rating            | 6.3 MW     | S6, S8  | Tender ceiling<br>(Höchstwert)  | 7.25 ct/kWh         | S2          | Loan term                          | 20 years                | S6, S9, S10 |
| Operating life            | 25 years   | S6, S14 | Futures 2027–29                 | 128 / 99 / 83 €/MWh | S12, S36    | Grace years                        | 3 years                 | S9, S10     |
| Financial close           | Jan 2027   | A       | Long-term power price           | 75 €/MWh            | S30 · A     | Repayment                          | Equal instalments (KfW) | S10         |
| Construction period       | 18 months  | S8      | Wind capture factor             | 0.80                | S11 · A     | Target DSCR (P50)                  | 1.20x                   | S26 · A     |
| ENERGY YIELD              |            |         | Direct-marketing fee            | 0.20 ct/kWh         | S6, S31 · A | Target DSCR (P90)                  | 1.00x                   | A           |
| Reference yield           | 3,623 h/yr | S6      | Municipal payment<br>(\$ 6 EEG) | 0.20 ct/kWh         | S3          | Maximum gearing                    | 80%                     | S6, S27     |
| Site quality (Gütefaktor) | 68%        | S7      | Premium settlement lag          | 3 months            | S5 · A      | Revenue the bank<br>counts on      | EEG floor               | S28 · A     |
| Availability              | 97.0%      | S20     | COSTS                           |                     |             | Debt service reserve               | 3 months                | S26 · A     |
| Degradation               | 0.20%/yr   | S21     | Total capex                     | 1,814 €/kW          | S6, S14     | TAX AND VALUATION                  |                         |             |
| Uncertainty, 1 year       | 12.8%      | S22 · A | Contingency                     | 3.0%                | A           | Legal form                         | GmbH & Co. KG           | S15, S16    |
| Output at negative prices | 5.0%       | S19 · A | Fixed opex, years 1–10          | 25.0 €/kW/yr        | S6          | Trade-tax multiplier<br>(Hebesatz) | 400%                    | S15 · A     |
|                           |            |         | Land lease                      | 10.0% of revenue    | S6, S23 · A | Depreciation period                | 16 years                | S17         |
|                           |            |         | Decommissioning cost            | 50 €/kW             | S24 · A     | Cost of equity                     | 8.0%                    | S6, S14     |
|                           |            |         | Generator grid fee              | 0.0 €/kW/yr         | S35         | Inflation from 2029                | 2.0%                    | S33         |
|                           |            |         |                                 |                     |             | Real WACC for the LCOE             | 3.9%                    | S14         |

S1–S36: sources listed on slide 18 · A: assumption — no public value for this case; see the methodology. All inputs, with hints and ranges, are in the calculator and the Excel workbook.

# 2,439 full-load hours at P50; in a bad year (one-year P90) the farm produces 84% of that

Full-load hours a year, before degradation



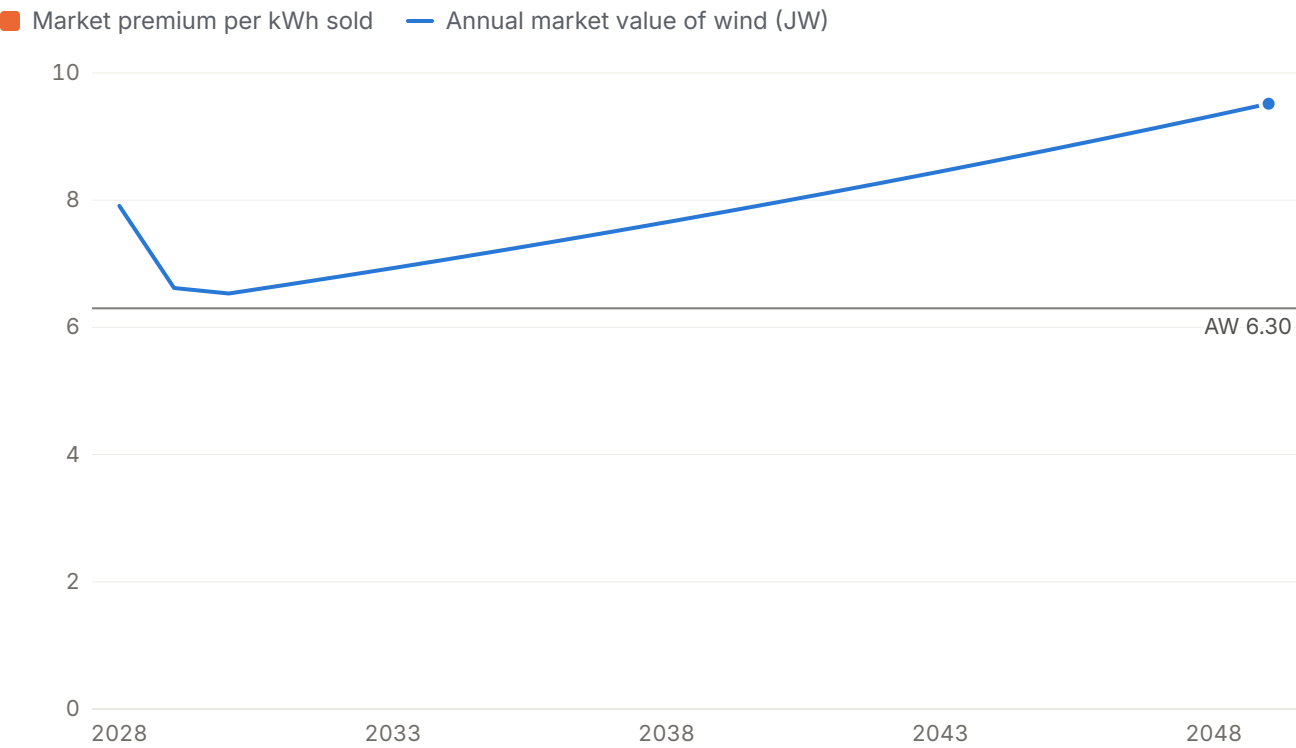
Output in 2029, the first full year (GWh)

|                                               |      |
|-----------------------------------------------|------|
| P50 output                                    | 76.7 |
| Sold: 5% falls into negative prices           | 72.8 |
| P90, one year (83.6% of P50)                  | 64.1 |
| P90, ten years (86.8% of P50)                 | 66.5 |
| P50 in 2052, after 0.2% a year of degradation | 73.2 |

- By law the site yield already excludes wake losses, up to 2% unavailability, electrical losses and curtailment required by the permit (Anlage 2 Nr. 7 EEG); the model deducts only availability below 98% and other losses.
- Site quality sets both the output and the correction factor of the AW: a better site produces more but receives less per kWh.
- $P90 = P50 \times (1 - 1.2816 \times \sigma)$ . The one-year value tests debt service in a bad year; the ten-year value describes a weaker wind resource.
- Grid curtailment under Redispatch 2.0 is compensated (§ 13a EnWG) and is not deducted.

# The AW of 6.30 ct/kWh stays below the expected market value in every year: the base case earns no premium

Base scenario during support, ct/kWh (nominal)



### How the premium is set

|                                                 |             |
|-------------------------------------------------|-------------|
| Award price (tender of 1 Aug 2026)              | 4.79 ct/kWh |
| × correction factor at 68% site quality (§ 36h) | 1.316       |
| = AW, 6.3036 rounded (§ 36h (5))                | 6.30 ct/kWh |
| Premium = max(0, AW – annual market value)      | Anlage 1    |
| No premium at negative prices: output share     | 5%          |
| Support: 20 years + § 51a extension             | +463 days   |
| Support ends                                    | 7 Oct 2049  |

### Premium by scenario

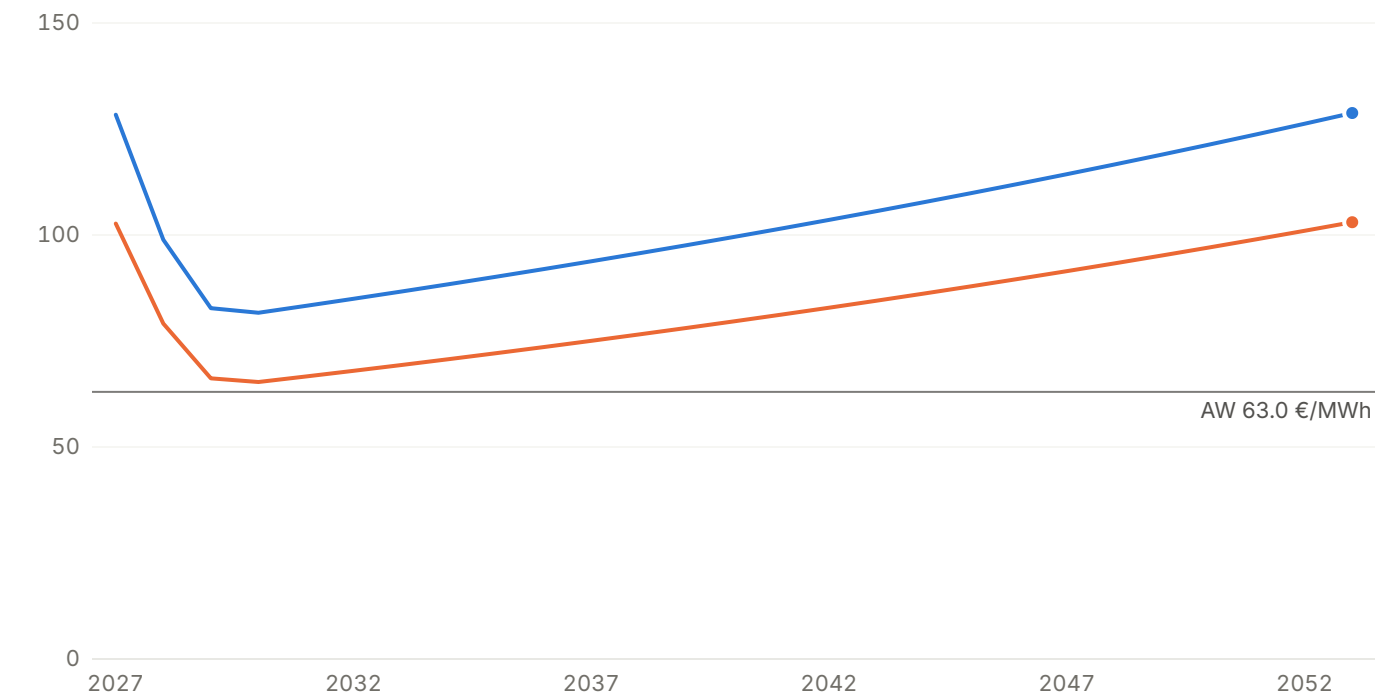
|           | Years paid | Total | AW from year 6 |
|-----------|------------|-------|----------------|
| Base      | 0 of 22    | €0.0m | 6.30 ct/kWh    |
| P90 1-yr  | 0 of 22    | €0.0m | 6.30 ct/kWh    |
| P90 10-yr | 1 of 22    | €0.4m | 6.80 ct/kWh    |
| Downside  | 15 of 22   | €8.4m | 6.80 ct/kWh    |

Advances follow the previous year's market value (§ 26); the rest arrives with the final settlement, assumed 3 months after the year end. The multi-year stresses re-determine the AW from year 6 (§ 36h (2)).

# Futures fall from 128 €/MWh (2027) to a long-term 75 €/MWh in 2026 money; wind earns 0.80 of the baseload price

Power prices, €/MWh (nominal)

— Baseload price — Market value of wind



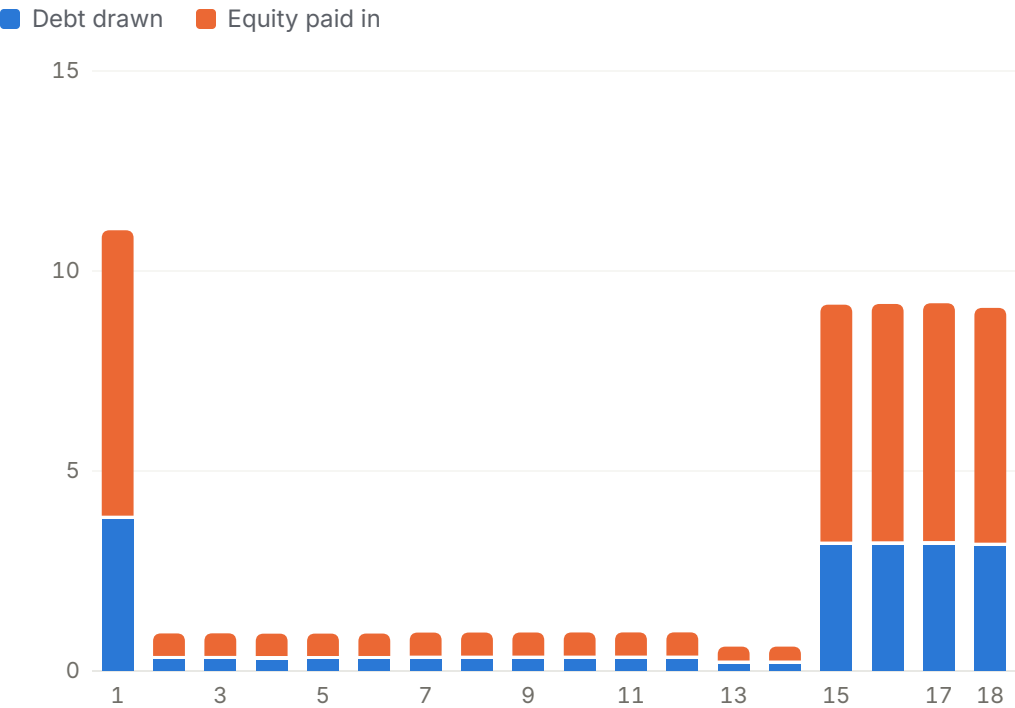
| Price inputs                                  |                          |
|-----------------------------------------------|--------------------------|
| EEX baseload futures 2027                     | 128.37 €/MWh             |
| EEX baseload futures 2028                     | 98.88 €/MWh              |
| EEX baseload futures 2029                     | 82.74 €/MWh              |
| Long-term baseload from 2030, 2026 money      | 75 €/MWh                 |
| Wind capture factor (market value ÷ baseload) | 0.80                     |
| Market value of wind in 2035 (nominal)        | 72.1 €/MWh               |
| Direct-marketing fee, 2026 money              | 0.20 ct/kWh              |
| After the EEG period                          | Market prices            |
| Inflation 2026/2027/2028, then                | 2.9% / 2.7% / 1.9%, 2.0% |

- The long-term price is the model's strongest assumption. Published scenarios range from about 65 to 101 €/MWh (Ariadne, Prognos, Agora).
- The farm's own capture price is assumed equal to the market value of all onshore wind; the 2023–2025 average factor was 0.81.
- Negative-price periods: 6.5% of the time, 5% of the output; the farm is curtailed then.

Construction and funding

Total uses of €59.3m — €57.1m capex, €1.1m financing costs, €1.1m reserves — funded with €20.6m of debt and €38.7m of equity

Funding by month after financial close, € million



Turbines: 10% at close, 70% on delivery, 20% at commissioning; civil works in thirds; development at close. VAT of 19% is bridged by a VAT loan for 2 months. Debt and equity are drawn pro rata.

Capex, net of VAT

|                                           | €/kW  | €m    |
|-------------------------------------------|-------|-------|
| Turbines incl. transport and installation | 1,163 | 36.64 |
| Foundations                               | 88    | 2.77  |
| Roads and crane pads                      | 119   | 3.75  |
| Grid connection                           | 133   | 4.19  |
| Development and permits                   | 163   | 5.14  |
| Compensation measures                     | 35    | 1.10  |
| Other                                     | 60    | 1.89  |
| Contingency 3%                            | 53    | 1.66  |
| Total capex                               | 1,814 | 57.14 |

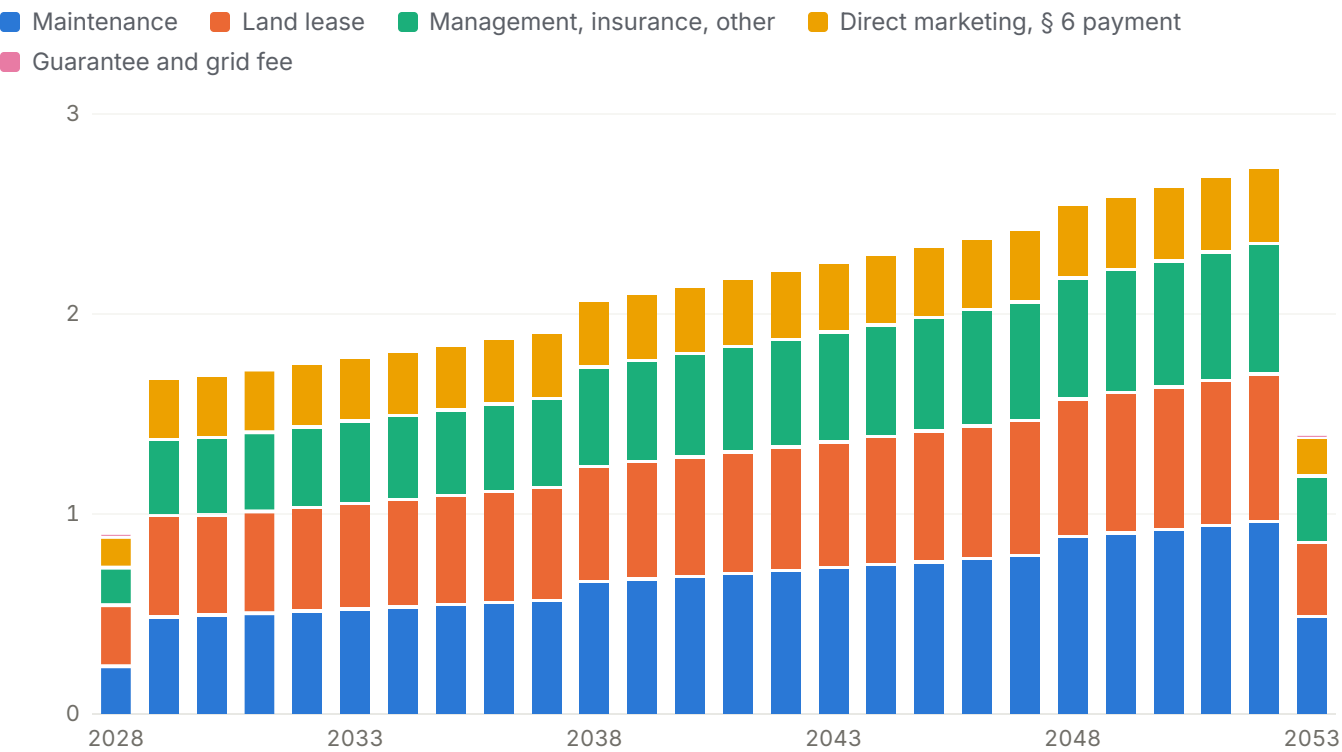
Sources and uses, € thousand

|                              |        |
|------------------------------|--------|
| Capex                        | 57,136 |
| Upfront fee                  | 207    |
| Commitment fees              | 216    |
| Interest during construction | 562    |
| VAT bridge interest          | 112    |
| Debt service reserve         | 573    |
| Start-up liquidity           | 498    |
| Total uses                   | 59,304 |
| Senior loan                  | 20,633 |
| Equity                       | 38,671 |
| Total sources                | 59,304 |

Gearing 34.8% · check: sources – uses = 0

# Operating costs of €1.7m in 2029, rising with inflation; decommissioning costs €2.7m at the end of life

Operating costs by year, € million (nominal)



Fixed items step up after years 10 and 20 (full-service maintenance ages). The first and last years are part years.

## 2029, the first full year

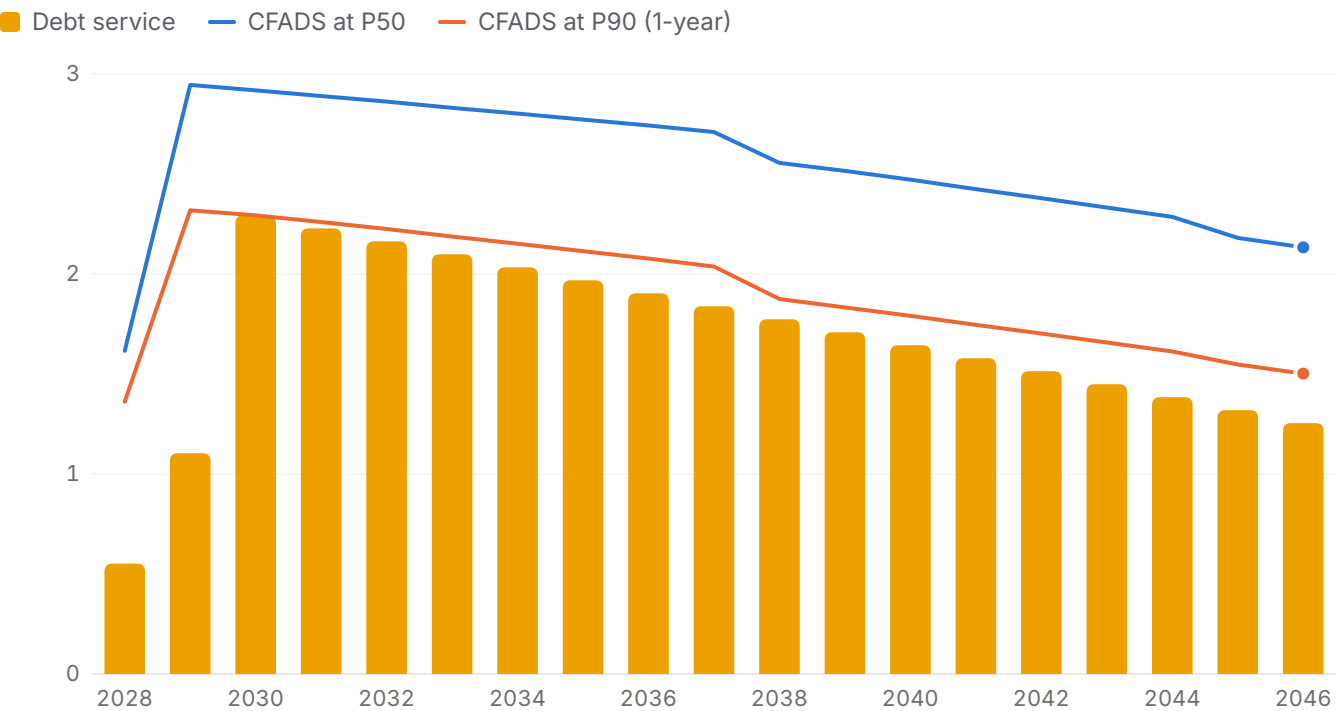
|                       | € thousand | €/MWh sold |
|-----------------------|------------|------------|
| Maintenance           | 484        | 6.65       |
| Management            | 138        | 1.90       |
| Insurance             | 35         | 0.47       |
| Other                 | 208        | 2.85       |
| Land lease            | 507        | 6.97       |
| Direct marketing      | 156        | 2.13       |
| Municipal payment § 6 | 153        | 2.11       |
| Guarantee fee         | 10         | 0.14       |
| Operating costs       | 1,691      | 23.22      |
| § 6 refund            | 0          | 0.00       |

## Decommissioning

|                                      |         |
|--------------------------------------|---------|
| Security: 159 m hub × €1,000 × 5     | €795k   |
| Guarantee fee 1.25% a year           | €10k    |
| Cost 50 €/kW (2026 money), paid 2053 | €2.70m  |
| Cash reserve built over the last     | 5 years |

# Debt of €20.6m (35% of uses) is limited by the P90 (1-year) DSCR target on the EEG floor

Lender’s case: revenue on the EEG floor, € million



The loan is the largest that keeps CFADS at least 1.20x × debt service at P50 and 1.00x × at P90 in every year of the lender’s case, and at most 80% of uses. Taxes, fees and the reserve depend on the loan: the model iterates.

Loan terms

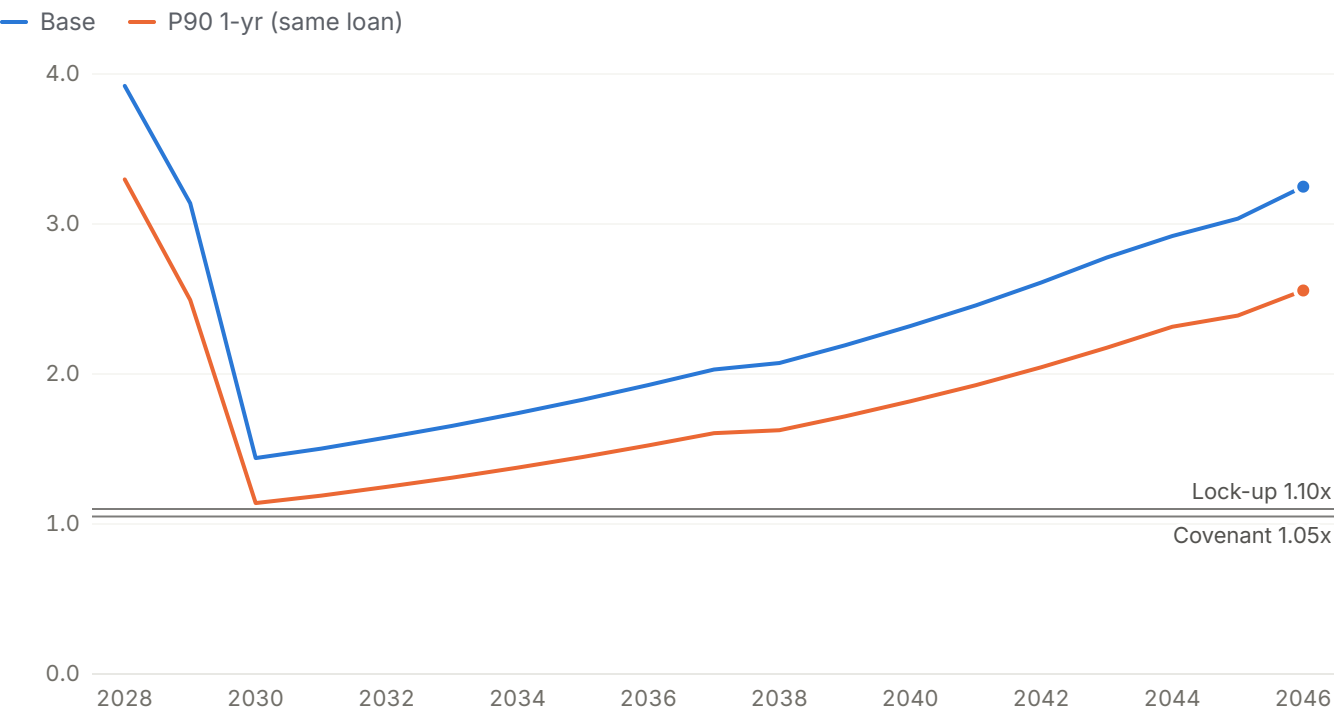
|                                         |                         |
|-----------------------------------------|-------------------------|
| Interest rate, fixed                    | 5.35%                   |
| Term from financial close / grace years | 20 / 3 years            |
| Repayment                               | Equal instalments (KfW) |
| Target DSCR P50 / P90 (1-year)          | 1.20x / 1.00x           |
| Revenue the bank counts on              | EEG floor               |
| Maximum gearing                         | 80%                     |
| Upfront / commitment fee                | 1.00% / 0.15% a month   |
| Debt service reserve                    | 3 months                |

Result

|                               |                                             |
|-------------------------------|---------------------------------------------|
| Loan                          | €20.63m                                     |
| Gearing (loan ÷ uses)         | 34.8%                                       |
| Limited by                    | P90 (1 - year) DSCR target on the EEG floor |
| Lowest lender DSCR, P50 / P90 | 1.27x / 1.00x                               |

# Lowest DSCR 1.44x in 2030 against a covenant of 1.05x; 1.14x in a one-year P90 stress

DSCR by year, operating cases with the base-case loan



DSCR = CFADS ÷ (interest + principal); CFADS = EBITDA – change in working capital – taxes. Grace years carry interest only, so their DSCR is high.

Cover and lock-up by scenario

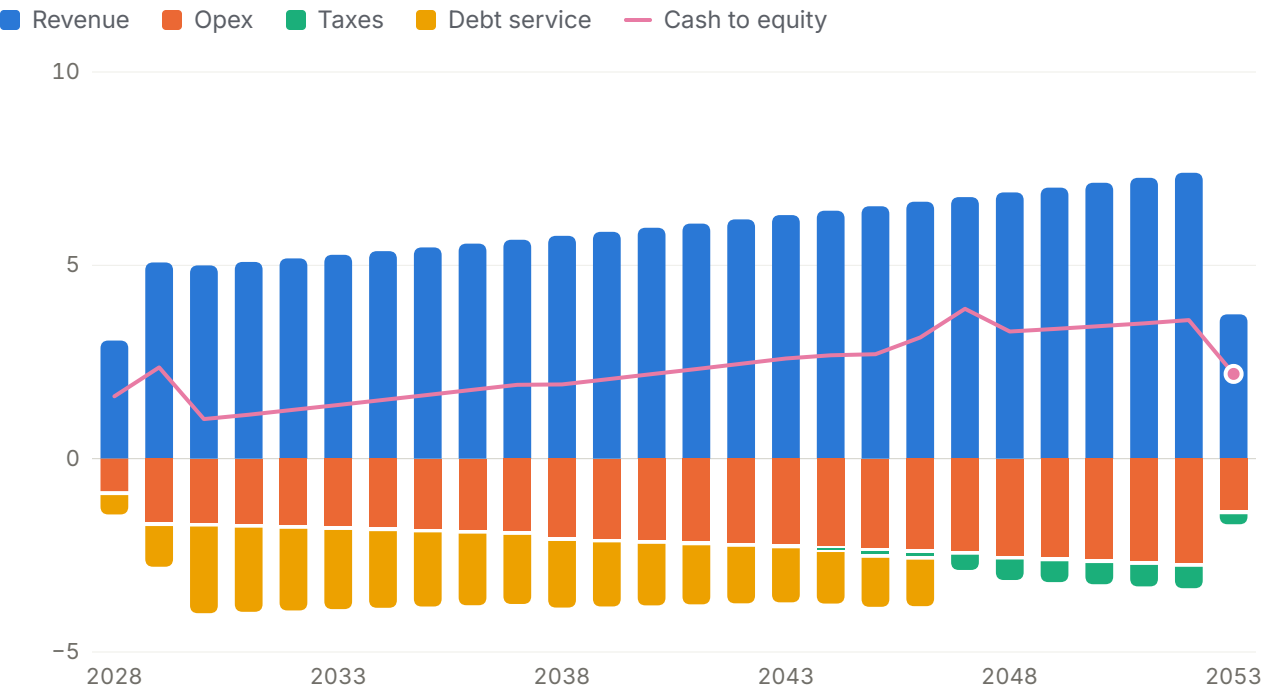
|           | Min DSCR | Year | Lock-up | LLCR  |
|-----------|----------|------|---------|-------|
| Base      | 1.44x    | 2030 | 0 yr    | 2.06x |
| P90 1-yr  | 1.14x    | 2030 | 0 yr    | 1.63x |
| P90 10-yr | 1.20x    | 2030 | 0 yr    | 1.73x |
| Downside  | 1.18x    | 2032 | 0 yr    | 1.48x |

Liquidity

|                                                |                |
|------------------------------------------------|----------------|
| Debt service reserve (3 months), funded at COD | €573k          |
| Start-up liquidity for first-year receivables  | €498k          |
| Average DSCR over the repayment years          | 2.20x          |
| Distributions held back below 1.10x            | 0 years        |
| Lowest cash balance                            | never negative |

# Tax is first due in 2044; over the life the owners receive €60.9m for €38.7m invested

Annual cash flows, € million (nominal)



A GmbH & Co. KG pays trade tax only (3.5% × 400% multiplier); its partners' income tax is not modelled. Straight-line depreciation over 16 years and loss carry-forwards keep taxable profit low in the early years.

Over the life, € million (nominal)

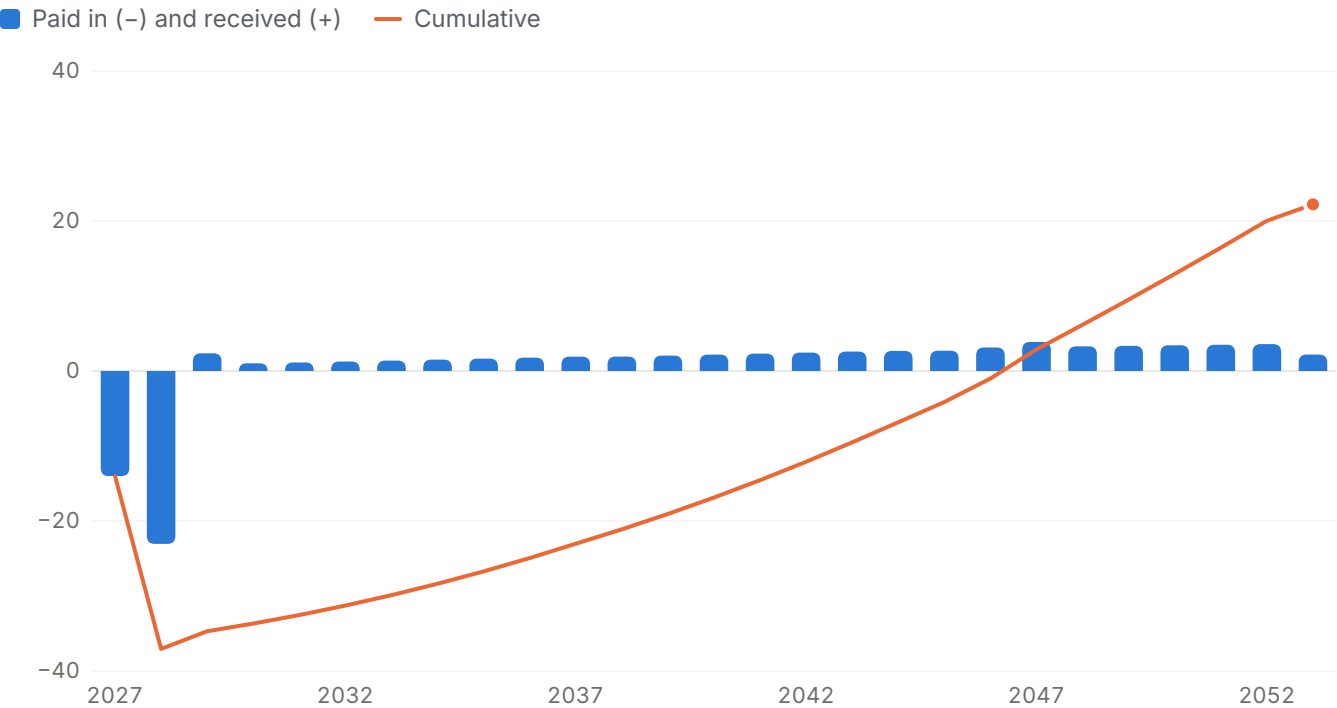
|                                          |             |
|------------------------------------------|-------------|
| Revenue                                  | 152.7       |
| Operating costs net of the § 6 refund    | -54.2       |
| <b>EBITDA</b>                            | <b>98.5</b> |
| Taxes                                    | -4.2        |
| Change in working capital                | 0.5         |
| <b>CFADS</b>                             | <b>94.8</b> |
| Interest                                 | -11.2       |
| Principal                                | -20.6       |
| Decommissioning paid                     | -2.7        |
| Reserves released (debt service reserve) | 0.6         |
| <b>Cash available to equity</b>          | <b>60.9</b> |
| Equity paid in during construction       | -38.7       |

Cash available to equity is before legal limits on distributions: § 30 GmbHG for a GmbH, § 172 (4) HGB for the limited partners of a KG. The model's book equity is negative in 2043–2050: a sign that such limits could bite, not a legal test.

Returns

Equity IRR 3.1% and NPV –€16.1m at 8%; the owners are paid back after 18.8 years

Owners' cash flows by year, € million (nominal)



Returns use the exact dates: monthly contributions during construction, cash to equity on 31 December (XIRR and XNPV, Actual/365).

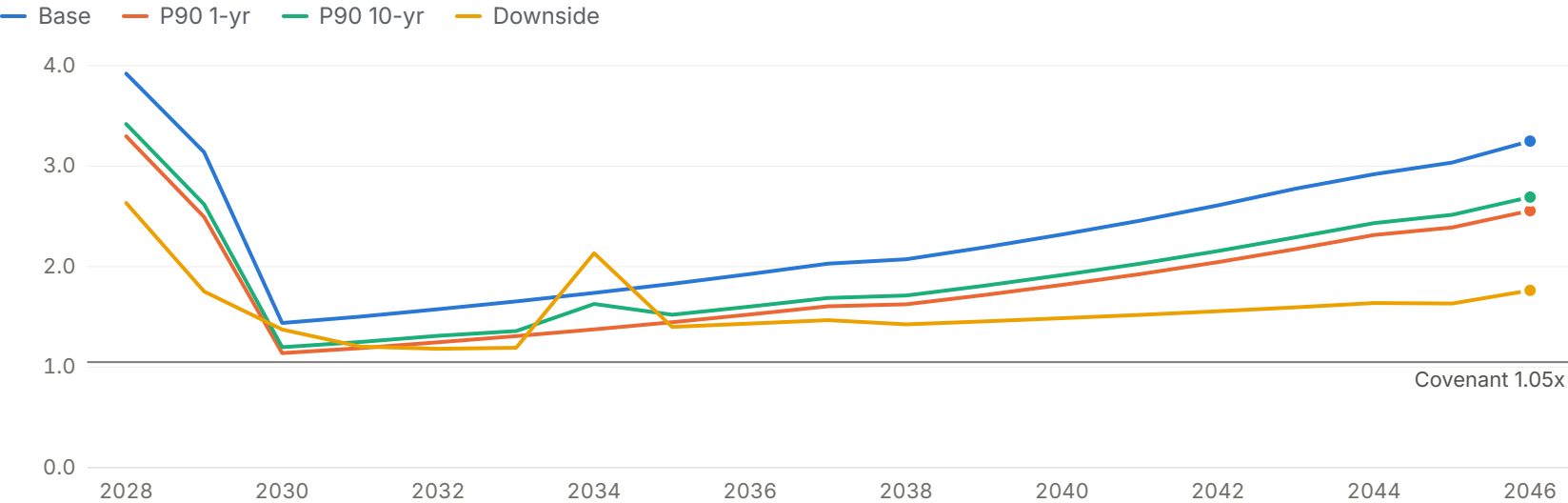
Returns, base case

|                                             |               |
|---------------------------------------------|---------------|
| Equity IRR (KG, before the partners' taxes) | 3.09%         |
| Project IRR, pre-tax / after tax            | 4.04% / 3.67% |
| Equity NPV at 8%                            | –€16.1m       |
| Project NPV at 5.8% (nominal WACC)          | –€11.2m       |
| Payback after commissioning                 | 18.8 years    |
| LCOE, real 2026 money at 3.9%               | 7.25 ct/kWh   |
| Anzulegender Wert (the EEG floor)           | 6.30 ct/kWh   |

- Negative leverage: the equity IRR (3.09%) is below the project IRR after tax (3.67%) — the project earns less than the loan's 5.35%.
- The LCOE of 7.25 ct/kWh is above the AW of 6.30 ct/kWh: the EEG floor alone does not cover the cost; the case relies on market prices.

In the downside the equity IRR falls to –2.1%; the lowest DSCR is 1.18x

|                        | Base                                              | P90 1-yr                                                               | P90 10-yr                                                        | Downside                                                                                                  |
|------------------------|---------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| What changes           | P50 output, base prices and costs; sizes the loan | One-year P90 output in every year (83.6% of P50) — the lender's stress | Ten-year P90 output in every year (86.8%), with the § 36h review | Ten-year P90, power prices –20%, fixed opex and grid fee +10%, capex +5% paid by the owners, § 36h review |
| Equity IRR             | 3.09%                                             | 0.61%                                                                  | 1.16%                                                            | –2.12%                                                                                                    |
| Project IRR, after tax | 3.67%                                             | 1.79%                                                                  | 2.22%                                                            | 0.01%                                                                                                     |
| Equity NPV             | –€16.1m                                           | –€23.1m                                                                | –€21.5m                                                          | –€28.8m                                                                                                   |
| Min DSCR (year)        | 1.44x (2030)                                      | 1.14x (2030)                                                           | 1.20x (2030)                                                     | 1.18x (2032)                                                                                              |
| Status                 | Funded, covenant met                              | Funded, covenant met                                                   | Funded, covenant met                                             | Funded, covenant met                                                                                      |

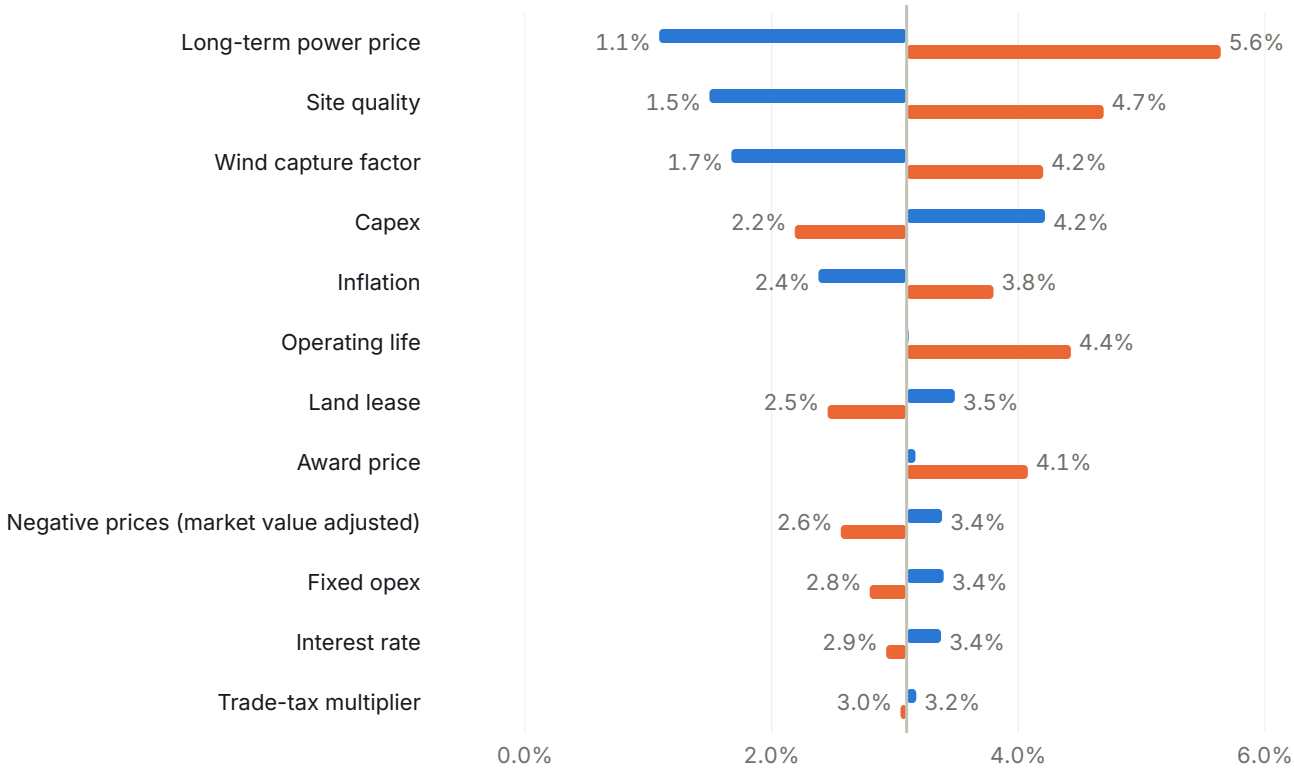


DSCR by year in each scenario. The stress scenarios keep the base-case loan — the lender’s view after financial close. Applying a P90 output to every year is a stress, not a 90% probability for the whole life.

# Long-term power price moves the equity IRR most: from 1.1% to 5.6%

Equity IRR, each driver at its low and high value

Low High



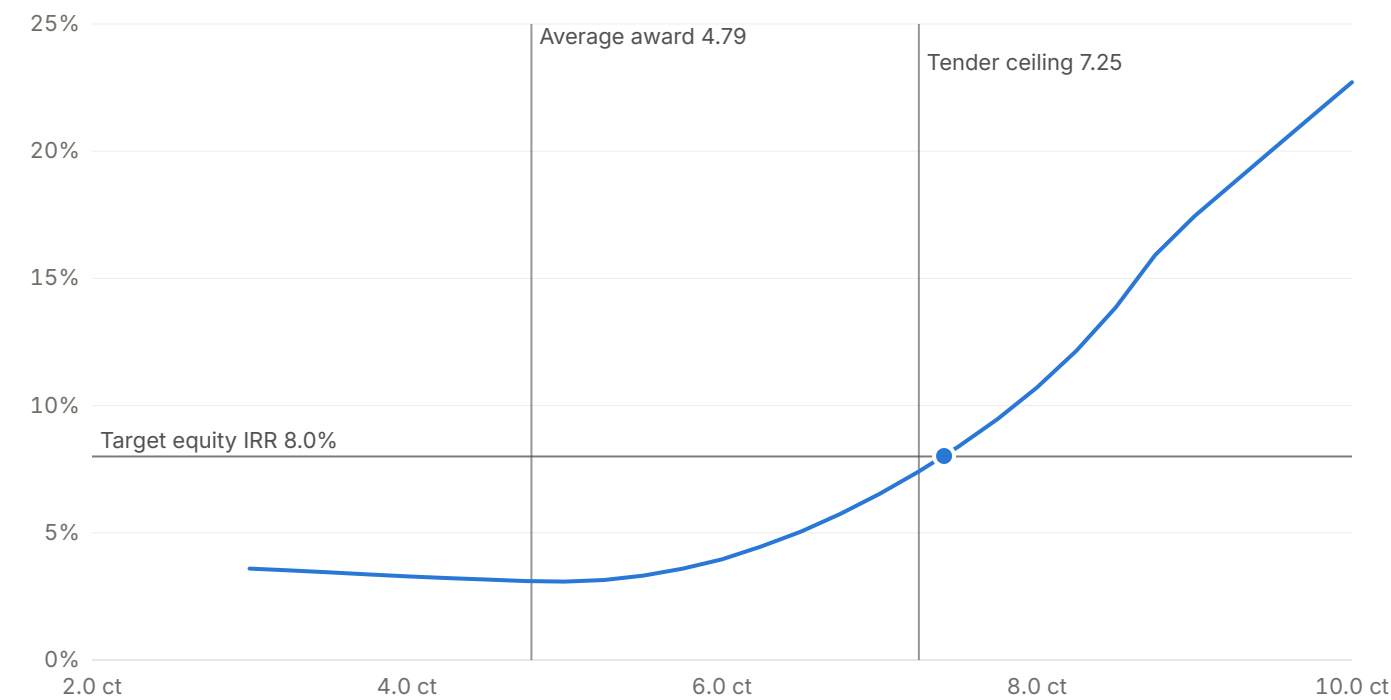
Base: 3.09%. Colour marks the side of the input (low / high), not better or worse.

| Settings                                |          |          |
|-----------------------------------------|----------|----------|
| Driver                                  | Low      | High     |
| Long-term power price                   | 60 €/MWh | 90 €/MWh |
| Site quality                            | 60%      | 76%      |
| Wind capture factor                     | 0.72     | 0.86     |
| Capex                                   | -10%     | +10%     |
| Inflation                               | -0.5 pp  | +0.5 pp  |
| Operating life                          | 25 yrs   | 30 yrs   |
| Land lease                              | 6%       | 14%      |
| Award price                             | 4.49 ct  | 6.06 ct  |
| Negative prices (market value adjusted) | 3%       | 9%       |
| Fixed opex                              | -10%     | +10%     |
| Interest rate                           | 4.40%    | 6.05%    |
| Trade-tax multiplier                    | 320%     | 450%     |

Each bar re-runs the model with the loan re-sized — the view before financial close. The negative-price driver also moves the capture factor with (1 – share). Drivers are sorted by the swing of the equity IRR.

# An 8% equity IRR needs an award of 7.41 ct/kWh — above the 7.25 ct/kWh ceiling

Equity IRR by award price, loan re-sized at every price



Search: a 0.25 ct grid, then bisection. Bids carry two decimals: the price shown is the lowest such price that qualifies.

Minimum award price

**7.41** ct/kWh

AW 9.75 ct/kWh at this site · target equity IRR 8.0%

- ✓ Target return reached: yes
- ✓ Financeable: fully funded, covenant met, award still valid (\$36e): yes
- ! Above the tender ceiling — such a bid would not be admitted.

|                                                   |             |
|---------------------------------------------------|-------------|
| Average award, 1 Aug 2026 (announced 17 Sep 2026) | 4.79 ct/kWh |
| Tender ceiling                                    | 7.25 ct/kWh |
| Award price in this case                          | 4.79 ct/kWh |

The IRR is not monotonic in the award price: a higher floor allows more debt at 5.35%, which can lower the equity return. The search therefore scans a 0.25 ct grid for the first qualifying price, bisects, and shows the lowest two-decimal price that qualifies.

## 8 risks you can test in the calculator, and 8 simplifications of the model to keep in mind

### Risks the calculator lets you test

- **Power price after the futures.** The long-term price is the strongest assumption; the tornado shows its weight.
- **Wind resource.** One-year and ten-year P90; a weaker site also re-sets the AW in the § 36h (2) reviews.
- **Negative prices.** No premium in those periods (§ 51 EEG); more of them also lower the capture factor.
- **EEG 2027 draft.** A two-sided premium is a switch — a simplified stress, not the draft's calculation; 18 banks warned of financing risks (Sep 2026).
- **Interest rate and loan terms.** KfW 270 at 5.35%; DSCR targets, gearing and the revenue the bank counts on are inputs.
- **Timing.** The award lapses 36 months after its announcement (§ 36e); a penalty applies after 30 months (§ 55). The checks flag late commissioning.
- **Generator grid fees.** Proposed in the regulator's AgNes process (4–7 €/kW a year); 0 in the base case, an input.
- **Distributions.** Cash to equity is before § 30 GmbHG and § 172 (4) HGB; negative book equity is flagged.

### What the model leaves out

- Operations are annual; output is spread evenly over the year (winter is in fact windier).
- The farm's capture price equals the market value of all onshore wind.
- The partners' income tax of a KG is not modelled; taxes are paid in the year they arise.
- One senior loan: no tranches, shareholder loans, refinancing or cash sweep; instalments during construction are not supported.
- Not modelled: the § 55 penalty for late commissioning, interest on § 36h paybacks, the interest barrier, and payments by the owners to cure a shortfall.
- Uncompensated grid curtailment (a draft of the grid package) and generator grid fees are not in the base case; the fee is available as an input.
- In the Excel workbook the loan, the total uses and the sculpted principal are solved on the website; the tornado and the bid calculator are included as values.
- The wind farm is fictional and the results are illustrative.

# 25 checks on every run, 36 public sources, a formula workbook checked in 40 variants

- Deterministic and nominal: the same inputs always give the same result. Construction runs monthly, operations by calendar year.
- Returns from dated cash flows (XIRR and XNPV, Actual/365). The loan is sized on the lender's case and iterated until it changes by less than €0.50.
- 25 checks in five groups on every run: calculation, funding, covenant, inputs and model scope.
- The Excel workbook rebuilds the model in formulas. Recalculated by Microsoft Excel in 40 variants, its roughly 26,000 formula cells agree with the engine: money to the cent.
- Every input has a public source, checked on 30 Sep 2026, or is a documented assumption.

Full methodology: [igorsabodakha.com/wind-farm-calculator/methodology/](https://igorsabodakha.com/wind-farm-calculator/methodology/)

Source code: [github.com/sabodaha/windpark-rechner](https://github.com/sabodaha/windpark-rechner)

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